Gr. 2 INTRODUCTION TO DEEP LEARNING Learning algorithms, Maximum likelihood estimation, Building machine learning algorithm, Neural Networks Multilayer Perceptron, Back-propagation algorithm and its variants Stochastic gradient decent, Curse of Dimensionality (15 questions)

Total points 14/15

The respondent's email (**192525391.simats@saveetha.com**) was recorded on submission of this form.

What do we call the special set of rules that a computer follows to learn how to solve a problem? *1/1

- A song
- A toy
- A story
- An algorithm

When a computer looks at lots of pictures of cats to learn what a cat looks like, what is it doing? *1/1

- Breaking the rules
- Learning from information
- Playing a game
- Taking a nap

How does a Neural Network work to solve a tricky puzzle? * 1/1

- [ ] It asks a human for the answer every time
- [ ] It uses one single giant eye
- [x] It connects different layers of ideas together like a brain
- [ ] It guesses randomly and never changes

Imagine a computer is trying to guess your favorite color. If it guesses wrong, what does it do next? * 1/1

- [ ] It says the same wrong answer again
- [x] It checks its mistake and tries to improve its next guess
- [ ] It cries because it is sad
- [ ] It gives up and turns off

What is the process called when a computer looks at its mistakes and sends a message back through its layers to fix them? *1/1

- (selected) Back-propagation ✓
- Front-running
- Side-stepping
- Speed-walking

If a computer wants to find the very best answer, how should it move toward the right solution? *1/1

- By jumping as far away as possible
- (selected) By taking small steps toward the right answer ✓
- By deleting all its work
- By staying in the same place forever

When a computer uses Stochastic Gradient Descent, it is like walking down a hill to find the bottom. How does it walk? * 1/1

- It rolls down as fast as it can without looking
- It flies to the top
- It takes many small, careful steps to reach the bottom ✓
- It walks in circles

Why do we want the computer to use Maximum Likelihood? * 1/1

- To find the guess that is the longest
- To find the guess that is the funniest
- To find the guess that is most likely to be correct ✓
- To find the guess that is the loudest

If you give a computer too many tiny details and choices at once, what usually happens? *0/1

- It becomes harder for the computer to find the right answer quickly
- It gets much faster
- It knows everything instantly
- It becomes a superhero

Correct answer

- It becomes harder for the computer to find the right answer quickly

What is the Curse of Dimensionality? * 1/1

- When a computer runs out of batteries
- Having too many choices or details that make a task confusing
- Having a magic wand that fixes everything

A Multilayer Perceptron is a type of Neural Network. What does it use to think? *1/1

- A metal shovel
- Only one single layer
- A box of crayons
- Many different layers of connected ideas

When we are Building a Machine Learning Algorithm, what are we actually making? *1/1

- A digital drawing of a tree
- A physical robot made of tin cans
- A new type of video game controller
- A smart way for the computer to learn on its own

If a computer is learning to recognize the letter A and it keeps calling it a B, what will back-propagation help it do? *1/1

- Erase its memory
- Ignore the letter A
- Change its internal connections to get closer to A next time
- Start guessing numbers instead

Why is it sometimes bad for a computer to have a cloud of too much data (the Curse of Dimensionality)? *1/1

- The computer turns into a real brain
- The computer becomes too smart for humans
- The computer might get lost in all the extra details
- The computer gets lonely

The computer becomes too smart for humans

The computer might get lost in all the extra details

The computer gets lonely

In Stochastic Gradient Descent, what does the computer do after every small step? *1/1

It forgets everything it just did

It takes a long nap

It adds more layers to its brain

It checks if it is closer to the best answer